

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

BOARD ORDER NO. 6-93-77
WDID NO. 6B150104001

REVISED WASTE DISCHARGE REQUIREMENTS

FOR

INYOKERN COMMUNITY SERVICES DISTRICT
WASTEWATER TREATMENT PLANT

Kern County

The California Regional Water Quality Control Board, Lahontan Region (Board) finds:

1. Discharger

Inyokern Community Services District submitted a complete Report of Waste Discharge (RWD) for the Inyokern Wastewater Treatment Plant on January 8, 1993. For the purposes of this Order, the Inyokern Community Services District is referred to hereinafter as the "Discharger."

2. Facility

The Inyokern Wastewater Treatment Plant is the Facility from which the discharge occurs. For the purposes of this Order, the Inyokern Wastewater Treatment Plant is referred to as the "Facility." The Facility discharges secondary domestic wastewater.

3. Permit History

The Board previously established Waste Discharge Requirements (WDRs) for the Discharger under Board Order No. 6-83-16, which was adopted on February 10, 1983. The Board also adopted Cease and Desist Order (CDO) No. 6-91-848 on July 11, 1991. The CDO found the Discharger to be in violation due to discharge of wastewater to undesignated disposal sites; not adequately protecting facilities from damage resulting from overflow, washout, or inundation from a storm or flood; and threatening to create a condition of nuisance and/or pollution. The CDO requires that compliance with the revised WDRs be achieved by July 1, 1995. Discharge Specifications violated in the CDO remain unchanged in this Board Order. Therefore, the CDO No. 6-91-848 and portions of Board Order No. 6-83-16 related to violations contained in CDO No. 6-91-848 remain in effect.

4. Reason for Action

These WDRs are being revised to reflect the upgrade and expansion of the existing treatment Facility.

5. Facility Location

The Facility is located in the Inyokern area within Sections 19, 29, 30, T26S, R39E, approximately 1.0 mile north of Inyokern, Kern County, in the China Lake Hydrologic Area of the Indian Wells Hydrologic Unit. The Facility location is shown on Attachments "A" and "B", which are made part of this Order. The Facility is located on land owned by the Discharger.

6. Description of Facility and Discharge

The Facility presently handles an average of 0.051 million gallons per day (mgd) of domestic wastewater with a design capacity of 0.070 mgd. A treatment process includes headworks, Imhoff tank, a Parshall flume with flow meter and chlorine contact tank, and percolation ponds. The Facility expansion will provide treatment for a daily average design flow of 0.150 mgd (peak flow of 0.375 mgd). The expanded Facility will include headworks, influent pump station, flow metering, lined facultative ponds operating in series, and rapid infiltration beds for secondary treated wastewater. The expansion is planned for completion by July 1, 1995.

7. Disposal Area

The existing wastewater treatment Facility uses percolation ponds for disposal. The Facility is designed to handle a maximum flow of 0.070 mgd but receive, on average approximately 0.05 mgd. The expanded Facility will use rapid infiltration beds for disposal. The rapid infiltration beds will be constructed at the original percolation pond location. The new Facility will be designed to handle a maximum flow of 0.15 mgd.

8. Sludge Disposal

Due to the nature of the treatment process, the removal of sludge will take place approximately every five years. Sludge will be disposed of at an authorized disposal site to be determined when the need arises.

9. Authorized Disposal Sites

The current percolation ponds and future infiltration beds are the only authorized wastewater disposal site. The percolation ponds are located on land owned by the Discharger.

10. Site Geology

The Inyokern area is generally sloping from south to north. Elevations range from about 2,500 feet above sea level at the south edge of the Discharger's boundary to 2,400 feet in the northeast. It is situated in an area of high heat flow and seismicity, characterized by northerly trending fault block mountains separated by alluvial valleys. The soil is made up of Helendale Loamy Sand which is a very deep and well drained soil on the alluvial fans and plains, except where weak cementation occurs. The permeability in the area is generally moderately slow.

11. Site Hydrology

The surface water of Inyokern is within the China Lake Hydrologic Area of the Indian Wells Hydrologic Unit which encompasses approximately 860 square miles. Over half of the Inyokern community is subject to flooding and is located within a flood hazard zone.

Ground water is the sole source of water within the Indian Wells Valley. Ground water extracted from the Inyokern area is generally of very high quality. Depth to ground water is 280 feet at the percolation ponds.

12. Receiving Waters

The receiving waters are the ground waters of the China Lake Hydrologic Area of the Indian Wells Hydrologic Unit.

13. South Lahontan Basin Plan

The Regional Board adopted a Water Quality Control Plan for the South Lahontan Basin on May 8, 1975, and this Order implements the Plan as amended.

14. Beneficial Uses

The beneficial uses of the ground waters of the China Lake Hydrologic Area of the Indian Wells Hydrologic Unit as set forth and defined in the Water Quality Control Plan for the South Lahontan Basin are:

- a. municipal and domestic supply
- b. agricultural supply
- c. industrial service supply
- d. freshwater replenishment

15. California Environmental Quality Act Compliance

The Discharger has certified a Negative Declaration dated December 1992, in accordance with the California Environmental Quality Act (Public Resources Code Section 21000 et seq.). The project, as approved by the Discharger, will not have a significant effect on water quality.

16. Notification of Interested Parties

The Board has notified the Discharger and interested parties of its intent to revise WDRs for the discharge.

17. Consideration of Public Comments

The Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the Discharger shall comply with the following:

I. DISCHARGE SPECIFICATIONS

A. Effluent Limitations

1. The current total effluent flow of wastewater during a 24-hour period shall not exceed 0.07 million gallons. The total effluent flow of wastewater during a 24-hour period after the Facility expansion shall not exceed 0.15 mgd.
2. The current maximum (peak) instantaneous flowrate of wastewater shall not exceed 0.15 mgd.

The maximum (peak) instantaneous flowrate of wastewater after the Facility expansion shall not exceed 0.375 mgd.

3. After the Facility expansion, wastewater made available to the authorized disposal sites shall not contain concentrations of parameters in excess of the following limits:

<u>Parameter</u>	<u>Units</u>	<u>Mean^{1/}</u>	<u>Maximum</u>
BOD ^{2/}	mg/l	30	45
Methylene Blue			
Active Substances	mg/l	1.0	2.0

4. All wastewater made available to the authorized disposal sites shall have a pH of not less than 6.0 pH units nor more than 9.0 pH units. A pH over 9.0 is allowed if it results from a biological process within the treatment plant.
5. All wastewater made available to the authorized disposal sites shall have a dissolved oxygen concentration not less than 1.0 mg/l.
6. All effluent made available for reclamation shall comply with the standard Department of Health Services Reclamation Requirements as specified in Chapter 3, Division 4, Title 22 of the California Code of Regulations.

B. Facultative Pond Lining Specifications

1. The facultative pond liner shall be designed and constructed in accordance with specifications outlined in Chapter 90, Section 93.422, of the publication "Recommended Standards for Wastewater Facilities," 1990 Edition^{2/}, a report prepared by the Great Lakes - Upper Mississippi River Board of State Public Health and Environmental Managers, or in accordance with other criteria protective of water quality and acceptable to the Board Executive Officer. The pertinent section of the referenced publication reads:

"Ponds shall be sealed such that seepage loss through the seal is as low as practicably possible. Seals consisting of soils, bentonite, or synthetic liners may be considered provided the permeability, durability, and integrity of the proposed material can be satisfactorily demonstrated for anticipated conditions. Results of a testing program which substantiates the adequacy of the proposed seal must be incorporated into and/or accompany the engineering report. Standard ASTM procedures or acceptable similar methods shall be used for all tests.

To achieve an adequate seal in systems using soil, bentonite, or other seal materials, the coefficient of permeability (k) in centimeters per second specified for the seal shall not exceed the value derived from the following expression:

$$k = 3.0 \times 10^{-9}L$$

Where L equals the thickness of the seal in centimeters. The "k" obtained by the above expression corresponds to a percolation rate of pond water of less than 500 gallons per day per acre (4.68m³/hd/d) at a water depth of six feet (1.8 m)."

C. Receiving Water Limitation

The discharge shall not cause the presence of the following substances or conditions in ground or surface waters of the Indian Wells Hydrologic Unit:

1. Any perceptible color, odor, taste or foaming.
2. Coliform organisms attributable to human wastes.
3. Toxic substances in concentrations that individually, collectively, or cumulatively cause detrimental physiological responses in human, plants, animals, or aquatic life.
4. Identifiable chlorinated hydrocarbons, organophosphates, carbamates, and other pesticide and herbicide groups, in summation, in excess of the lowest detectable.

D. General Requirements and Prohibitions

1. There shall be no discharge, bypass, or diversion of raw or partially treated sewage, sewage sludge, grease, or oils from the collection, transport, treatment, or disposal facilities to adjacent land areas or surface waters.
2. Surface flow or visible discharge of sewage or sewage effluent from the authorized disposal/reclamation sites to adjacent land areas or surface waters is prohibited.
3. The vertical distance between the liquid surface elevation and the lowest point of a pond dike or the invert of an overflow structure shall not be less than two (2.0) feet.
4. The discharge shall not cause a pollution as defined in Section 13050 of the California Water Code, or a threatened pollution.
5. Neither the treatment nor the discharge shall cause a nuisance as defined in Section 13050 of the California Water Code.
6. The discharge of wastewater except to the authorized disposal sites is prohibited.
7. The integrity of the facultative pond liners shall be maintained throughout the life of the ponds and shall not be diminished as the result of any maintenance of cleaning operation.

8. Prior to any removal of sludge, the Discharger shall notify the Board of the removal and shall submit a sludge removal plan to the Board 90 days prior to any such removal for approval by the Executive Officer. The Discharger shall also notify the Kern County Environmental Health Services Department for any sludge removed for process or disposal within Kern County. The Discharger shall comply with all existing federal and state laws and regulations that apply to sewage sludge use and disposal practices. The Discharger shall further comply with all requirements regarding sludge use and disposal specified in the Clean Water Act Section 405(d).

II. PROVISIONS

A. Rescission of Waste Discharge Requirements

All portions of Board Order No. 6-83-16 except which are necessary to implement CDO No. 6-91-848 are hereby rescinded. In cases where portions of Board Order No. 6-83-16 vary from the corresponding portions in this Order, this Order shall have precedence.

B. Operator Certificates

The Discharger's wastewater treatment plant shall be supervised by persons processing a wastewater treatment plant operator certificate of appropriate grade pursuant to Chapter 3, Subchapter 14, Title 23, California Code of Regulations.

C. Standard Provisions

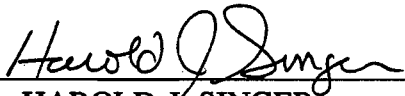
The Discharger shall comply with the "Standard Provisions for Waste Discharge Requirements," dated July 1, 1993, in Attachment "C", which is made part of this Order.

D. Monitoring and Reporting

1. Pursuant to the California Water Code Section 13267(b), the Discharger shall comply with the Monitoring and Reporting Program No. 93-77 as specified by the Executive Officer.

2. The Discharger shall comply with the "General Provisions for Monitoring and Reporting," dated July 1, 1993, which is attached to and made part of the Monitoring and Reporting Program.

I, Harold J. Singer, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Lahontan Region, on August 12, 1993.



HAROLD J. SINGER
EXECUTIVE OFFICER

- Attachments:
- A. Regional Location Map
 - B. Site Location Map
 - C. Standard Provisions for Waste Discharge Requirements

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- 1/ The arithmetic mean of lab results for effluent samples collected in a period of 30 consecutive days.
 - 2/ Biochemical Oxygen Demand (5 day, 20°C) of a filtered sample. Using a No. 1 Whatman filter or equivalent and reseeded with an unfiltered sample.
 - 3/ This report is published by Health Education Services, P.O. Box 7126, Albany, NY 12224; telephone number 518-439-7286.

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

MONITORING AND REPORTING PROGRAM NO. 93-77
WDID NO. 6B150104001

FOR
INYOKERN COMMUNITY SERVICES DISTRICT

INYOKERN
WASTEWATER TREATMENT PLANT

Kern County

I. MONITORING

A. Flow Monitoring

The following shall be monitored.

1. The total volume, in million gallons, of wastewater flow to the treatment plant for each day.
2. The total volume, in million gallons, of wastewater flow to the treatment plant for each month.
3. The average flow rate, in million gallons per day (mgd), of wastewater to the treatment plant calculated for each month.
4. The maximum instantaneous flow rate, in mgd, of wastewater to the treatment plants that occurs each day.
5. The freeboard (distance from the top of the lowest part of the dike to the wastewater surface in a pond) measured each month in each pond. If the pond does not contain wastewater, indicate that it is empty.

B. Plant Effluent Monitoring

1. Beginning immediately, grab samples of the oxidation pond effluent shall be collected and analyzed to determine the magnitude of the following parameters, to be reported quarterly:

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
BOD ^{1/}	mg/l	Monthly
COD ^{2/}	mg/l	Monthly
pH	pH units	Monthly

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Methylene Blue		
Active Substances	mg/l	Monthly
Oil & Grease	mg/l	Monthly
Total Dissolved		
Solids (TDS)	mg/l	Monthly
Dissolved Oxygen	mg/l	Monthly
Kjeldahl Nitrogen	mg/l as N	Semiannually
Nitrate Nitrogen	mg/l as N	Semiannually
Ammonia Nitrogen	mg/l as N	Semiannually
Chloride	mg/l	Semiannually
Sodium	mg/l	Semiannually
Sulfate	mg/l	Semiannually
Total Hardness	mg/l as CaCO ₃	Semiannually
Boron	mg/l	Semiannually
Fluoride	mg/l	Semiannually

C. Ground Water Monitoring

1. By April 1, 1995, the Discharger shall install additional ground water monitoring wells in order to comply with the following requirements.

The disposal site monitoring system, at a minimum, shall include the following:

- a. A minimum of three monitoring wells shall be installed to determine the gradient of the ground water at the disposal site.
- b. Additional wells shall be installed, if necessary, to ensure that at least one (1) well is upgradient and two (2) are downgradient of the wastewater disposal site. The Discharger shall demonstrate that at least two downgradient wells are located such that ground water potentially impacted by the Discharger will be monitored.
- c. The specific design and location of the wells shall be reviewed and approved by the Executive Officer.
- d. An as-built design report shall be submitted within 60 days after the ground water monitoring system is installed. This shall include a Statement of Certification signed by a California Registered Civil Engineer, or Engineering Geologist regarding the placement, lithology, and construction of the wells.

2. Grab samples from the entire thickness or the upper 20 feet, whichever is less, of the uppermost ground water bearing zone shall be collected from the disposal site monitoring wells and analyzed to determine the magnitude of the following parameters:

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Methylene Blue		
Active Substances	mg/l	Annually ^{3/}
Total Dissolved Solids	mg/l	Annually ^{3/}
Kjeldahl Nitrogen	mg/l as N	Annually ^{3/}
Nitrate Nitrogen	mg/l as N	Annually ^{3/}
Ammonia Nitrogen	mg/l as N	Annually ^{3/}
Chloride	mg/l	Annually ^{3/}
Purgeable Halocarbons ^{4/5/}	mg/l	Annually
Base/Neutral/Acid		
Extractable Organics ^{4/6/}	mg/l	Annually
Purgeable Aromatic ^{4/7/}	mg/l	Annually
Arsenic	mg/l	Annually
Silver	mg/l	Annually
Cadmium	mg/l	Annually
Copper	mg/l	Annually
Chromium	mg/l	Annually
Lead	mg/l	Annually
Mercury	mg/l	Annually
Molybdenum	mg/l	Annually
Nickel	mg/l	Annually
Selenium	mg/l	Annually
Zinc	mg/l	Annually

- a. The depth to ground water in each well shall be measured and recorded each time a well is sampled.
- b. The velocity and direction of ground water flow under the Facility site shall be determined at least annually unless it can be shown that no changes have occurred. A graphical presentation of the direction of ground water flow shall be submitted annually.

D. Pretreatment

1. A summary of activities undertaken to monitor and regulate industrial wastewater sources shall be prepared.

- a. an inventory of significant industrial users and any users of grease traps, including names, addresses, categories, industrial pollutants, and volumes. A significant industrial user is either:
 - (1) an industrial user discharging more than 25,000 gallons per day to the treatment plant,
 - (2) is a categorical industrial user as defined in 40 CFR 400-471, or
 - (3) can pass through or cause interference to the wastewater treatment plant.
 - b. A discussion of upset, interference, or pass through incidents, if any, at the treatment plant which the Discharger knows or suspects was caused by industrial users.
 - c. A discussion of enforcement actions taken or proposed.
2. A summary of the pretreatment functions including, but not limited to:
- a. necessary legal authorities,
 - b. pretreatment requirements, and
 - c. status of funding and personnel to implement the pretreatment program.

II. REPORTING

A. General Provisions

The Discharger shall comply with the "General Provisions for Monitoring and Reporting" dated July 1, 1993, which is attached to and made part of this Monitoring and Reporting Program.

B. Submittal Periods

1. Quarterly monitoring reports containing the above required information shall be submitted to the Board beginning on October 15, 1993. Subsequent monitoring reports shall be submitted to the Board by the 15th day following the end of each quarter.

2. Annual monitoring reports shall be submitted along with the monthly monitoring report submitted to the Board by the 15th day following the end of the year.

Ordered by:


HAROLD J. SINGER
EXECUTIVE OFFICER

Dated: August 12, 1993

Attachments: General Provisions for Monitoring and Reporting

- 1/ Biochemical Oxygen Demand (5 day, 20°C) of a filtered sample. Using a No. 1 Whatman filter or equivalent and reseeded with an unfiltered sample.
- 2/ Chemical Oxygen Demand of a filtered sample.
- 3/ Quarterly samples for the first year and annually thereafter.
- 4/ Analysis shall be conducted for those substances included on the EPA list of priority pollutants and all other substances known to be discharged to the California City Area sewer system.
- 5/ Purgeable organics analysis shall be conducted in accordance with EPA method 601, 40 CFR part 136, Appendix A.
- 6/ Base/neutral/acid extractable organics analysis shall be conducted in accordance with EPA method 625, 40 CFR, part 136, Appendix A.
- 7/ Purgeable Aromatics analysis shall be conducted in accordance with EPA method 602, 40 CFR, part 136, Appendix A.